

Yurou Liu

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RESEARCH INTEREST

LLM4Science — building large language models and generative foundation models that speak the language of science, spanning scientific generative modeling, cross-modal representation learning, and developing scalable agent systems for complex scientific workflows.

EDUCATION

Renmin University of China Ph.D. Student in Artificial Intelligence	Beijing <i>Sept. 2023 – Present</i>
Beijing Normal University B.Sc. in Computer Science	Beijing <i>Sept. 2019 – June 2023</i>

INTERNSHIP EXPERIENCE

Tongyi Lab, Alibaba Group Research Intern on LLM4Science (Mentor: Mingyang Li)	Beijing <i>Dec. 2024 – Present</i>
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PUBLICATIONS

* denotes equal / core contribution.

Speaking the Language of Science: Toward a General-Purpose Generative Foundation Model for the Natural Sciences Yurou Liu*, Mingyang Li*, Jieping Ye, Bing Su, Ji-Rong Wen, Zheng Wang	Tongyi Tech Report 2026
DrugTrail: Interpretable Drug Discovery via Structured Reasoning and Druggability-Tailored Preference Optimization Yurou Liu, Mingyang Li, Xinyuan Zhu, Rui Jiao, Yiming Dong, Xinyu Tang, Yang Liu, Jieping Ye, Bing Su, Zheng Wang	ICLR 2026
Towards High Data Efficiency in Reinforcement Learning with Verifiable Reward Xinyu Tang*, Zhenduo Zhang*, Yurou Liu*, Wayne Xin Zhao, Zujie Wen, Zhiqiang Zhang, Jun Zhou	ICLR 2026
DenoiseVAE: Learning Molecule-Adaptive Noise Distributions for Denoising-based 3D Molecular Pre-training Yurou Liu*, Jiahao Chen*, Rui Jiao*, Jiangmeng Li, Wenbing Huang, Bing Su	ICLR 2025
Probing RLVR Training Instability through the Lens of Objective-Level Hacking Yiming Dong, Kun Fu, Haoyu Li, Xinyuan Zhu, Yurou Liu, Lijing Shao, Jieping Ye, Zheng Wang	ICML 2026
Learning 3D Anisotropic Noise Distributions Improves Molecular Force Field Modeling Xixian Liu, Rui Jiao, Zhiyuan Liu, Yurou Liu, Yang Liu, Ziheng Lu, Wenbing Huang, Yang Zhang, Yixin Cao	NeurIPS 2025